

XII: Chemistry 1st-Term Chemistry Paper - 2081 (2024)

Grade XII: Chemistry

Date:

MCQs

Set 'A'

Group 'A'

Page:

Solutions Chemistry

By Pankaj Purbey

1. b. 50 ml

2. c. 49

3. c. NH_4Cl & NH_4OH

4. a. 6.63

5. b. +3

6. a. $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot 4\text{H}_2\text{O}$

7. d. ZnSO_4

8. a. anhydrous $\text{ZnCl}_2 + (\text{conc. HCl})$

9. b. DDT

10. c. $\text{CH}_3\text{COOCH}_2\text{CH}_3$

11. d. nitronium ion

MCQs

Group 'A'

Set 'B'

1. c. $\text{CH}_3\text{COOCH}_2\text{CH}_3$

2. a. anhydrous $\text{ZnCl}_2 + (\text{conc. HCl})$

3. b. DDT

4. d. nitronium ion

5. a. 6.63

6. c. NH_4Cl & NH_4OH

7. c. 49

8. b. 50 ml

9. a. $[\text{Cu}(\text{NH}_3)_4]\text{SO}_4 \cdot 4\text{H}_2\text{O}$

10. b. +3

11. d. ZnSO_4

Q.N.1. (a) Ans: Law of Chemical Equivalence states that "ratio of masses in which two reactants react is in the ratio of their equivalent masses."

$$\frac{\text{Mass of reactant A}}{\text{Equivalent mass of A}} = \frac{\text{Mass of reactant B}}{\text{Equivalent mass of B}}$$

2nd Part:

We have,

$$\text{Normality (N)} = \frac{\text{No. of gram equivalent}}{\text{Volume of solution in ml}} \times 1000 \quad \text{--- (1)}$$

$$\text{No. of gram equivalent} = \frac{\text{Total weight}}{\text{Equivalent weight}}$$

$$= \frac{W}{E} \quad \text{--- (2)}$$

From eqn (1) & (2),

$$\text{Normality (N)} = \frac{W}{E} \times \frac{1}{\text{Volume of solution in ml}} \times 1000$$

$$\text{Or, } N = \frac{W}{E} \times \frac{1}{V_{ml}} \times 1000$$

$$\text{Or, } \frac{W}{E} = \frac{N V_{ml}}{1000}$$

for acid,

$$\text{No. of gram equivalent} = \left(\frac{N_{\text{Vme}}}{1000} \right)_{\text{acid}}$$

for base,

$$\text{No. of gram equivalent} = \left(\frac{N_{\text{Vme}}}{1000} \right)_{\text{base}}$$

Now,

$$\text{No. of gram equivalent of acid} = \text{No. of gram equivalent of base}$$

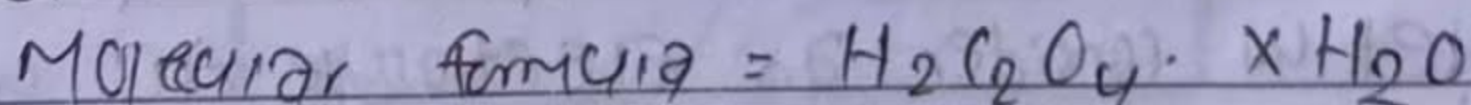
$$\text{Or, } \left(\frac{N_{\text{Vme}}}{1000} \right)_{\text{acid}} = \left(\frac{N_{\text{Vme}}}{1000} \right)_{\text{base}}$$

$$\therefore N_a V_a = N_b V_b$$

Hence, eqn (ii) is required eqn. for normality.

1. (b) solution:

Let no. of water of crystallization be x .



$$\text{Molecular wt.} = 90 + 18x$$

for oxalic acid crystals

$$\text{Weight (w)} = 7.875 \text{ gm}$$

$$\text{Equivalent wt. (E)} = \frac{\text{Mol. wt.}}{2}$$

$$= \frac{90 + 18x}{2}$$

$$= 45 + 9x$$

for NaOH

$$\text{Normality (N)} = 0.2 \text{ N}$$

16 ml of dilute solution requires

40 ml of 0.2 N NaOH

Then, 250 ml of dilute

$$\text{solution requires } \frac{40}{16} \times 250 = 625 \text{ ml}$$

$$\therefore V = 625 \text{ ml}$$

Now,

$$\frac{W}{E} = \frac{NV}{1000}$$

$$\text{Or, } \frac{7.875}{45+9x} = \frac{0.2 \times 625}{1000}$$

$$\text{Or, } \frac{7.875 \times 1000}{45+9x} = 125$$

$$\text{Or, } \frac{7875}{45+9x} = 125$$

$$\text{Or, } 63 = 45+9x$$

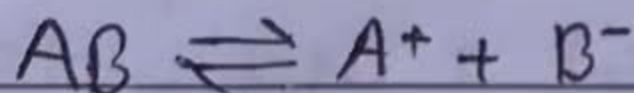
$$\text{Or, } 9x = 18$$

$$\therefore x = 2 //$$

Hence, number of water of crystallization is 2. //

Q.N.2. (a) Ans:

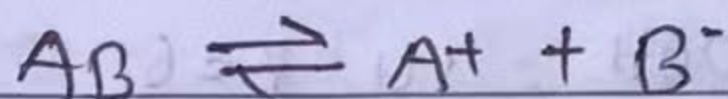
Let us consider an electrolyte AB which ionizes as:



Dissociation constant or ionization constant (K_{eq}) can be calculated as,

$$K_{eq} = \frac{[A^+][B^-]}{[AB]} \dots (1)$$

If initial concentration of an electrolyte AB is 'c' mole/lit and α be degree of ionization, then concentration term for different species,



Initially $c \quad 0 \quad 0$

At equilibrium $c - c\alpha \quad c\alpha \quad c\alpha$

Substituting these values in eqn (i),

$$K_{eq} = \frac{c\alpha \times c\alpha}{c - c\alpha} = \frac{c\alpha^2}{1 - \alpha} \quad \text{--- (ii)}$$

For weak electrolyte, $\alpha \ll 1$. So, $1 - \alpha \approx 1$.

Then, eqn (ii) becomes,

$$K_{eq} = c\alpha^2$$

$$\text{Or, } \alpha^2 = \frac{K_{eq}}{c}$$

$$\therefore \alpha = \sqrt{\frac{K_{eq}}{c}}$$

This is the required equation for Ostwald's dilution law for weak electrolyte.

2nd part: Limitation:

For strong electrolyte, $\alpha = 1$.

$$K_{eq} = \frac{c\alpha^2}{1 - \alpha} = \frac{c\alpha^2}{1 - 1} = \frac{c\alpha^2}{0} = \infty \text{ which}$$

is not possible. Hence, Ostwald's dilution law is only applicable for weak electrolytes.

Q.N.2. (b) solution:

Given,

$$pH \text{ of } KOH = 8$$

$$\therefore pOH = 14 - 8 = 6$$

$$\text{Also, } [OH^-] = 10^{-6} M$$

When it is diluted 1000 times,

$$N_1 = 10^{-6} N \quad N_2 = ?$$

$$V_1 = x$$

$$V_2 = 1000x$$

we have,

$$N_1 V_1 = N_2 V_2$$

$$\therefore N_2 = \frac{N_1 V_1}{V_2} = \frac{10^{-6} \times x}{1000x} = 10^{-9} N$$

Also,

$$\text{Total } [OH^-] = [OH^-] \text{ from } KOH + [OH^-] \text{ from } H_2O$$

$$= 10^{-9} + 10^{-7}$$

$$= 1.01 \times 10^{-7}$$

Then,

$$pOH = -\log (1.01 \times 10^{-7}) = 6.995$$

Now,

$$pH = 14 - 6.995 = 7.004 //$$

Hence, the pH of diluted base is 7.004 //

Q.N.3. (a) Ans: The effect of temperature and catalyst in reaction rate:

Temperature: When temperature of reactant molecules increase, K.E. of reactant molecules increases. This increase in K.E. results in the increase of effective collision. As a result, rate of reaction increases.

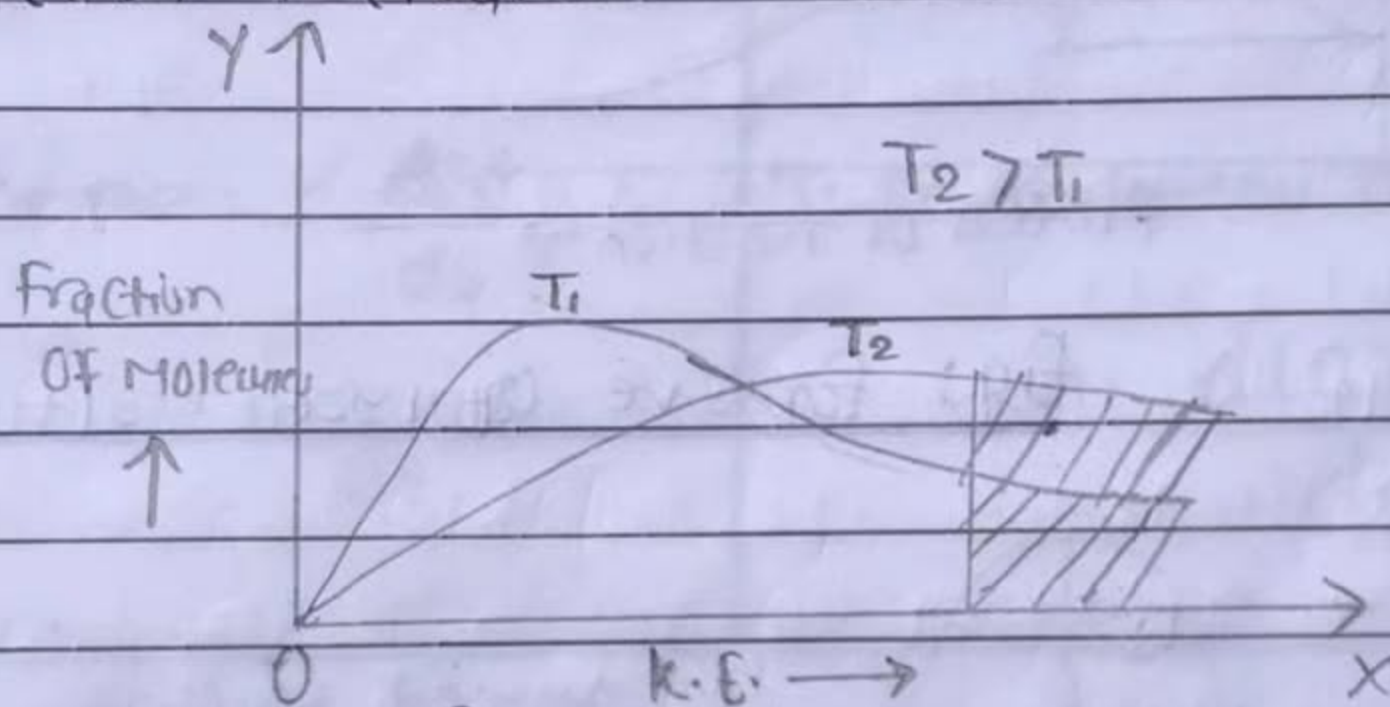


Fig: Maxwell distribution of energy curve

Catalysts:

Catalyst are the substances that alter rate of reaction by providing alternative path for chemical reaction without changing initial and final position. 2 types of catalyst are there:

Positive Catalyst: Positive catalyst provides alternative path with lower activation energy and hence large fraction of molecules cross energy barrier increasing rate of reaction:

Negative Catalyst: Negative catalyst provide alternative path with higher activation energy and hence fewer fraction of molecules cross energy

barrier decreasing rate of reaction.

Energy profile diagram:

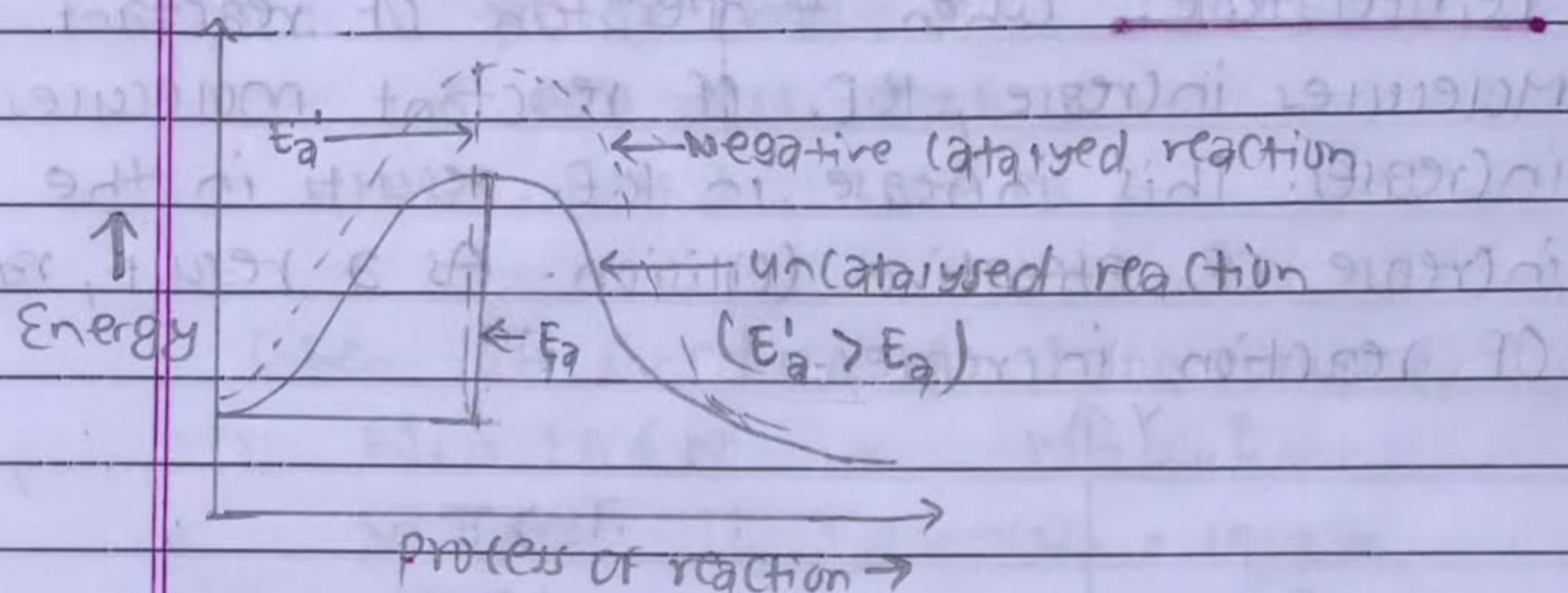


fig: for -ve catalyzed reaction

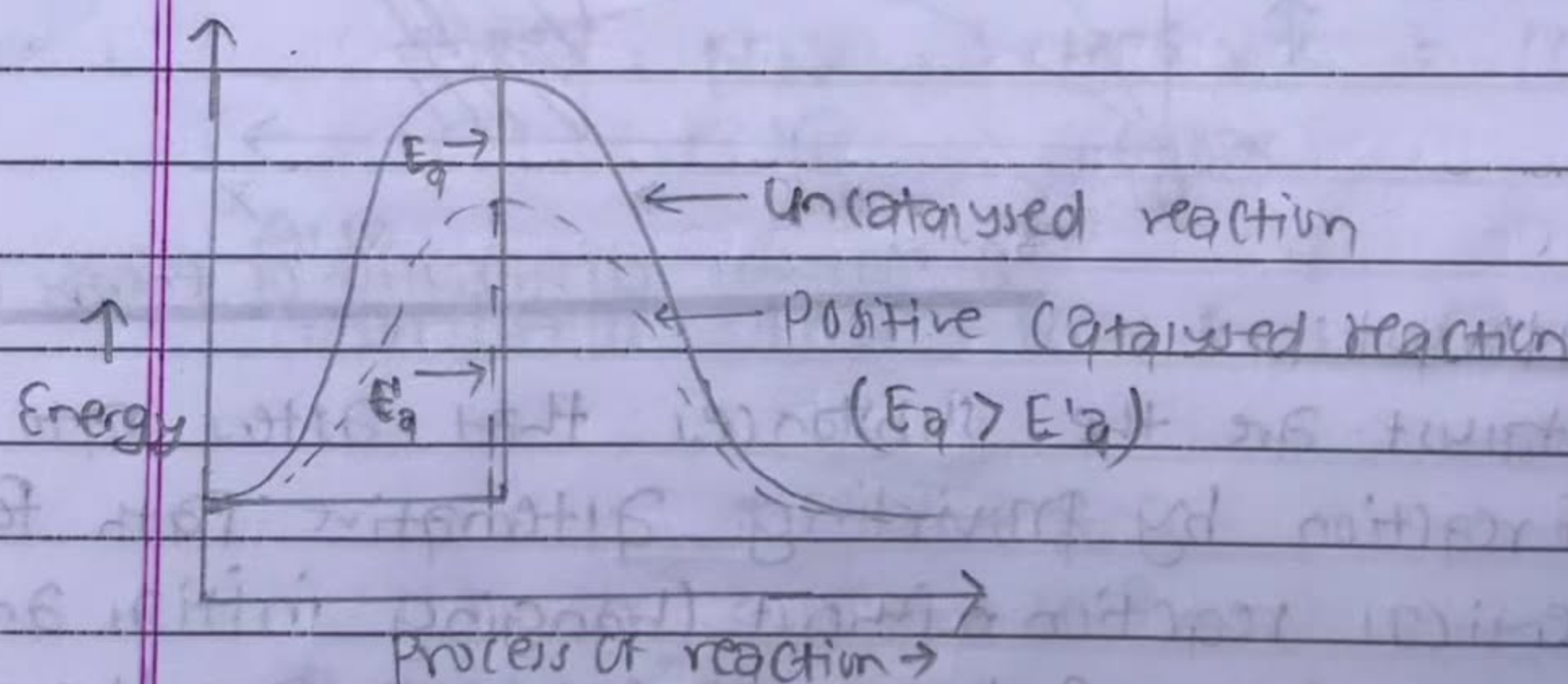


fig: for +ve catalyzed reaction

A.N.3.

(b) Ans: The differences between average rate of reaction and instantaneous rate of reaction are:

Average rate

1. It is the rate of reaction during certain interval of time.

2. It is finite and measurable.

$$3. \text{ Average rate} = - \frac{\Delta[A]}{\Delta t}$$

$$\text{or, } + \frac{\Delta[B]}{\Delta t}$$

Instantaneous rate

1. It is the rate of reaction during particular instant of time.

2. It is non finite and non measurable.

$$3. \text{ Instantaneous rate} = - \frac{d[A]}{dt}$$

$$\text{or, } \frac{d[B]}{dt}$$

Q.N.U. (2) Ans: 'Extraction of Zinc'

The extraction of zinc from zinc blende (ZnS) is carried out by carbon reduction process which involves following steps:

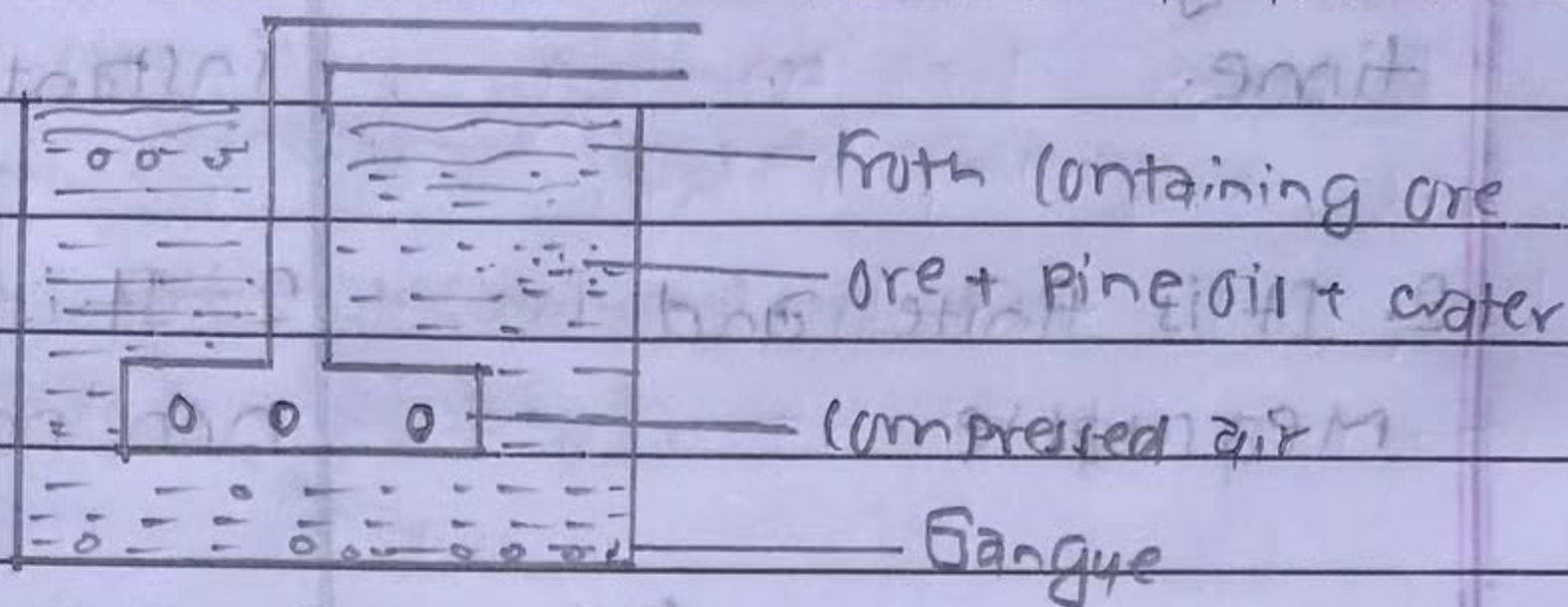
(i) Crushing and Pulverization:

Big lumps of ore are crushed and pulverized into fine powder by Jaw crusher and big mills.

(ii) Concentration:

It is concentrated by froth floatation process when crushed sulphide ore is mixed

With water and Pine Oil in tank is agitated by blowing air. Impurities settle at bottom of tank while ore come to top as froth.

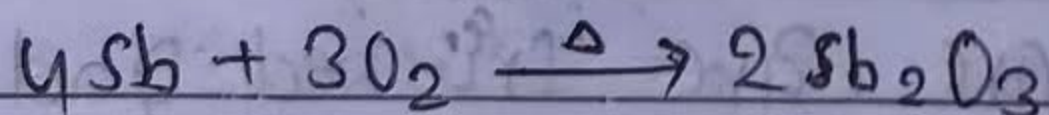
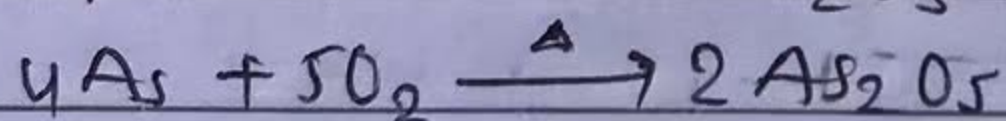


Eg: Froth floatation

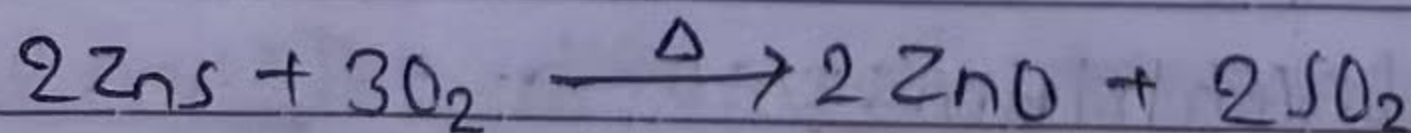
Roasting:

Concentrated ore is heated in presence of air at about 900°C in a reverberating furnace.

- Volatile impurities and moisture are removed.
- Impurities like P_4 , As , Sb , S etc. get oxidized.

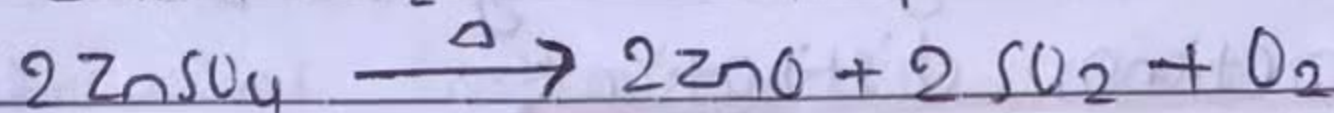
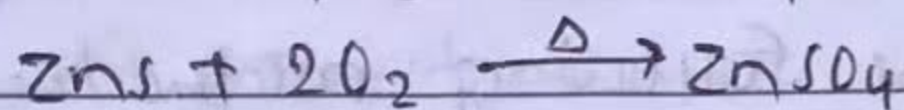


- Sulphide ore gets converted into oxide ore.



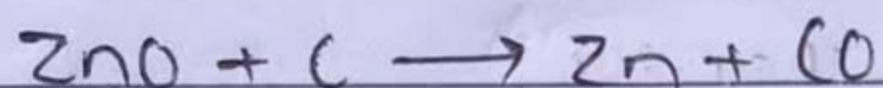
- Roasting is done above 900°C , sulphate

is formed and decomposed to zinc oxide.



(iv) Reduction:

Zinc oxide obtained by roasting is heated with carbon in furnace to reduce oxide into metallic form.



Vertical Retort Process:

Mixture of ZnO and coke in 2:1 in form of small briquets are fed from charging door of vertical furnace. Furnace is heated by producer gas at 1400°C , ZnO is reduced to zinc and then vaporizes. Thus, obtain zinc in condenser after release of volatile impurities called as spelter zinc.

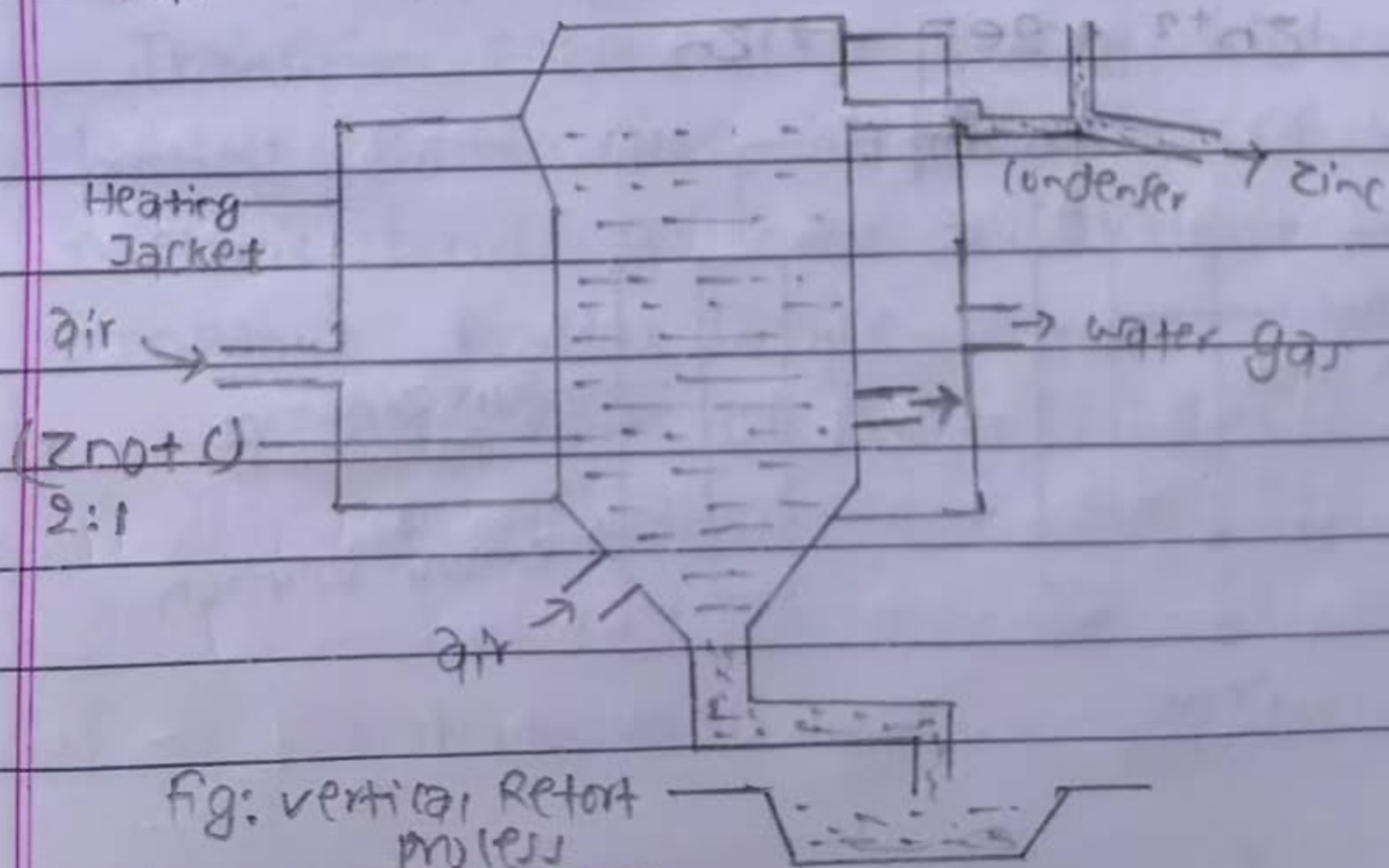


Fig: vertical Retort process

① Purification: Impurities are removed from crude zinc by following methods.

(a) Fractional distillation:

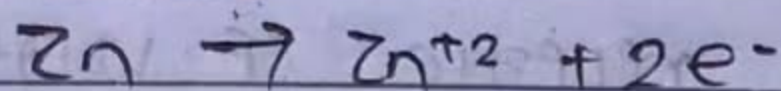
When distillation is carried out about 1000°C , Zn and Cd are distilled out leaving behind Fe and Pb. After redistilled at 800°C , Cd is distilled out whereas Zn metal remains in distillation flask.

(b) Electrolytic refining:

Anode as spelter zinc, Cathode as pure zinc, electrolytic solution is zinc sulphate.

Pure zinc is deposited at cathode and impurities are left behind at bottom of anode as anode mud.

At anode:



At cathode:

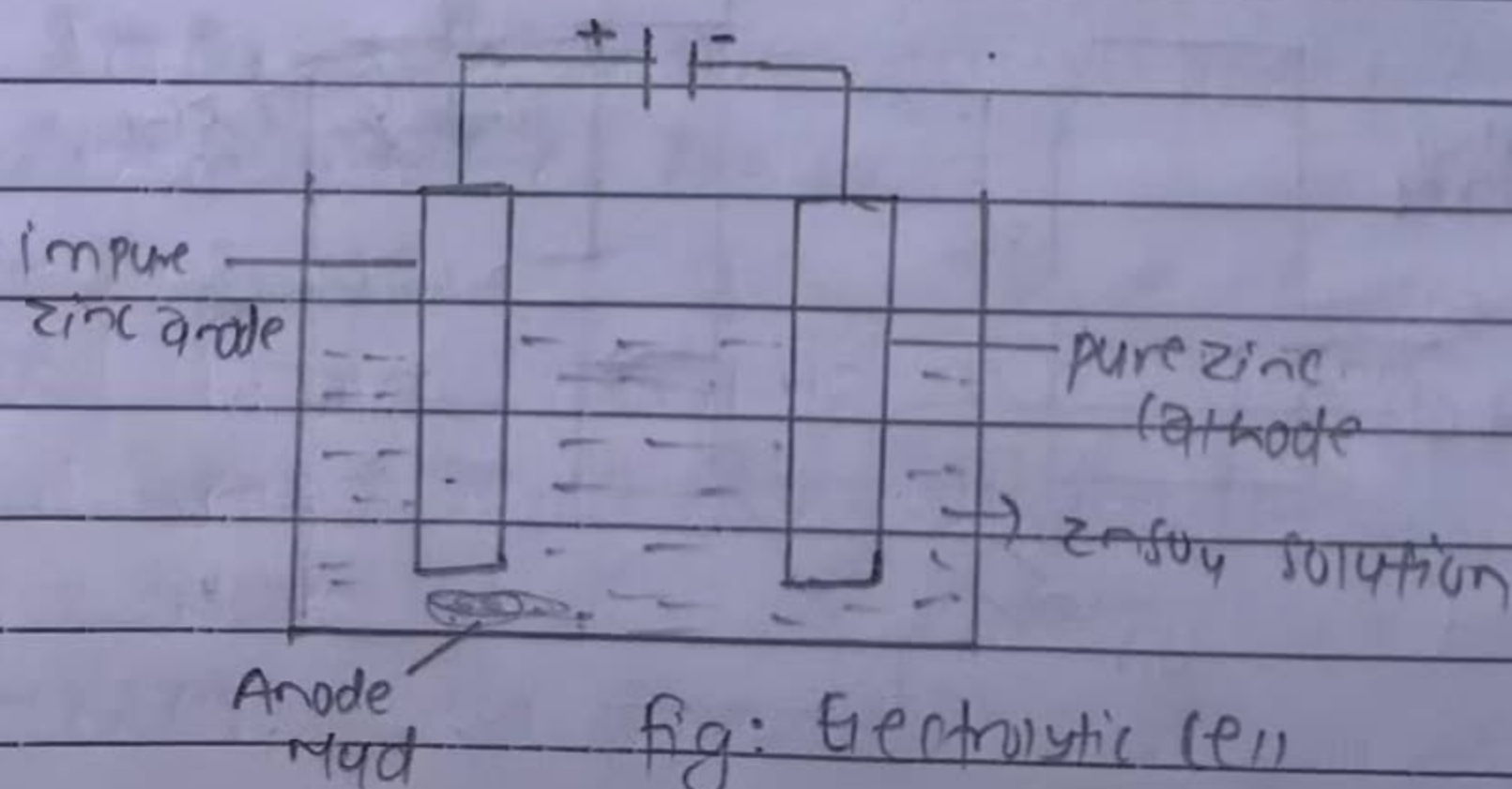
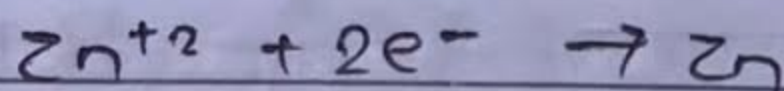
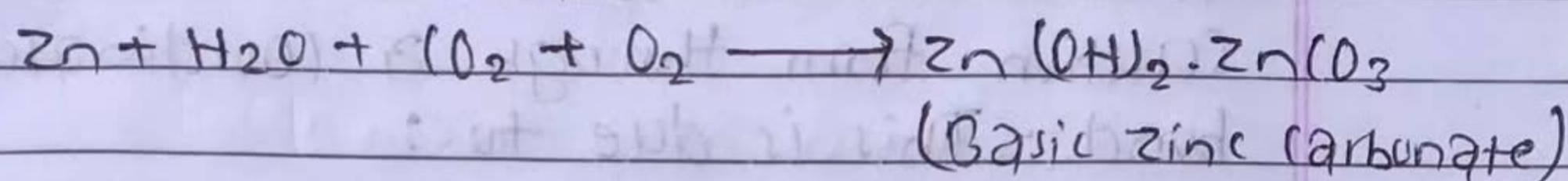


Fig: Electrolytic cell

Q. (b) Ans: When Zinc is exposed to moist air, it gets tarnished due to formation of basic zinc carbonate.



Q. N.S. Ans: The main general characteristics of Transition metals with relevant examples are:

① Variable valency and Oxidation States:

It is due to presence of comparable energy having $(n-1)d$ and ns orbital in their valence shell. Involvement of ns orbital electrons form lower oxidation state while both $(n-1)d$ and ns orbital electrons form higher oxidation state.

for eg: ④ Fe +2, +3 ⑤ Cu +1, +2

② High melting and boiling Points:

Transition metals show very high melting and boiling point due to presence of very strong metallic bond as well as covalent bond in their metallic crystal lattice.

for eg: M.P. of Cu = 1083°C

B.P. of Cu = 2567°C

→ Copper

(III) Coloured Compounds:

Transition metals have a high tendency to form coloured compounds in solid or solution form through a coordinate covalent bond. This is due to:

- * Small size of atoms and ions of transition metals
- * high nuclear or ionic charge
- * availability of vacant d-orbitals of suitable energy for bonding.

for eg: Tetrammine copper(II) ion $[\text{Cu}(\text{NH}_3)_4]^{2+}$ is blue.

(IV) Magnetic properties:

Many compounds of transition elements exhibit magnetic properties. This is due to presence of unpaired electrons in d-orbitals and show paramagnetic properties. The complexes are diamagnetic substances if their d-orbital electrons are paired.

For eg: Fe^{3+} are more paramagnetic than Fe^{2+} .

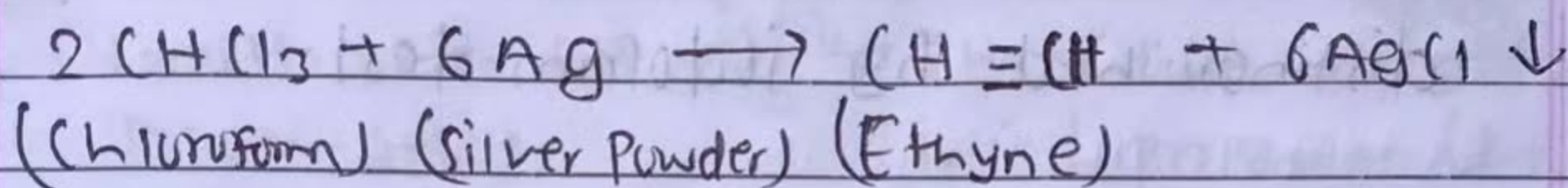
(V) Catalytic Property:

Catalytic property is due to the presence of partially filled (n-d) orbital.

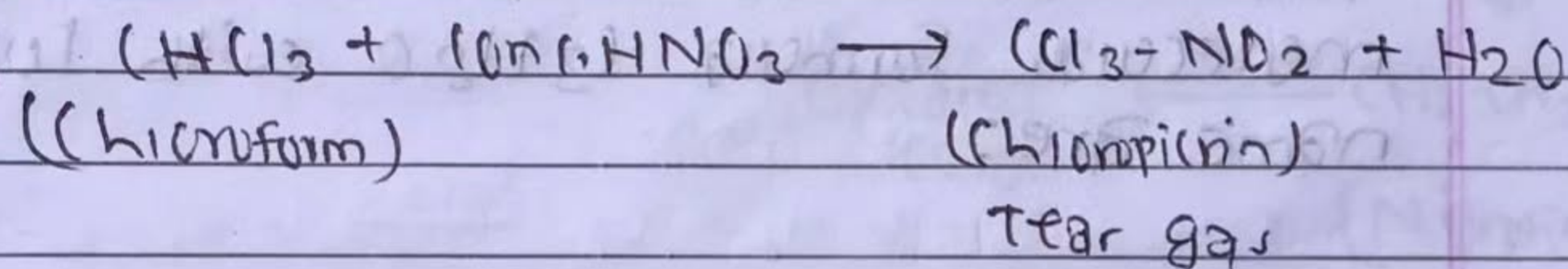
For eg: Fe is used as catalyst in Haber's process.

Q.N. 6. (a) Ans: Starting from trichloromethane
 $(CHCl_3)$

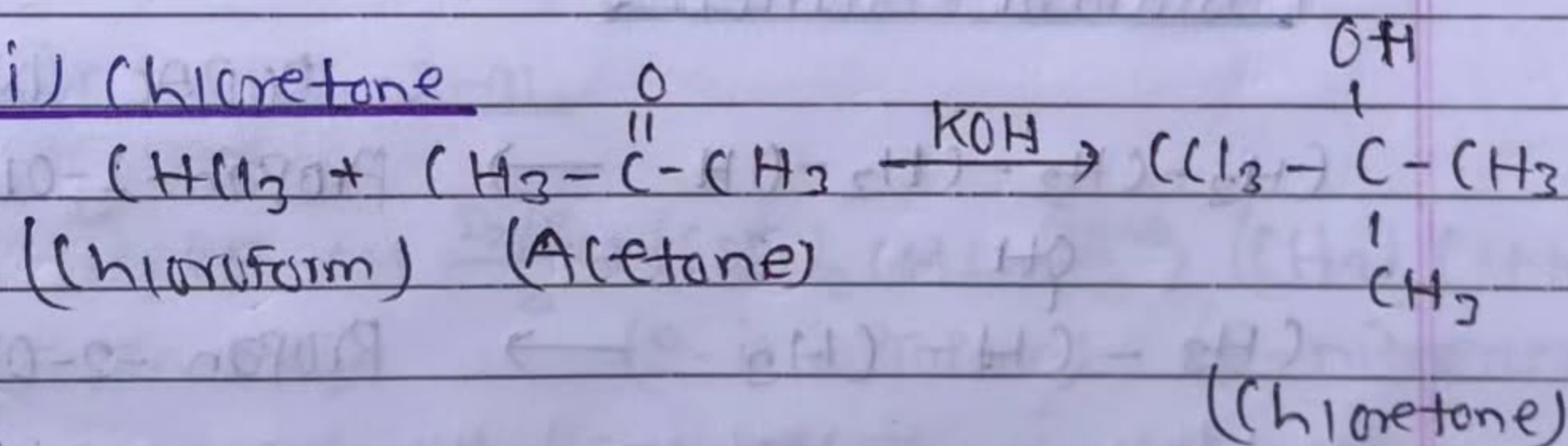
(i) Ethyne



(ii) Chloropicrin

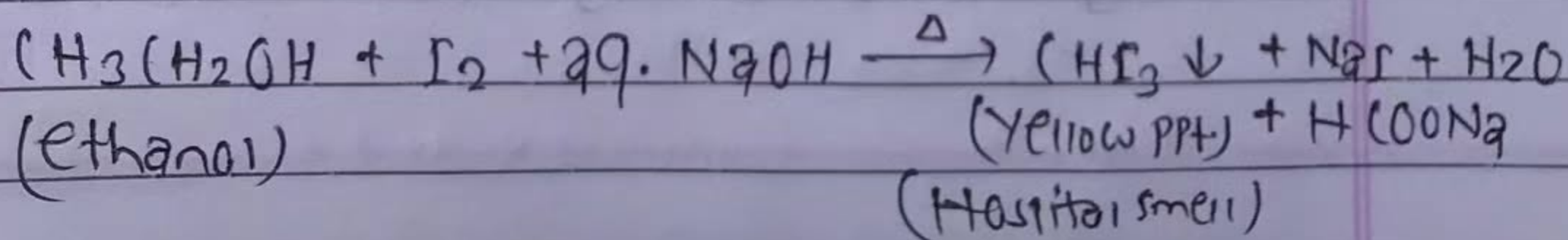


(iii) Chloroacetone

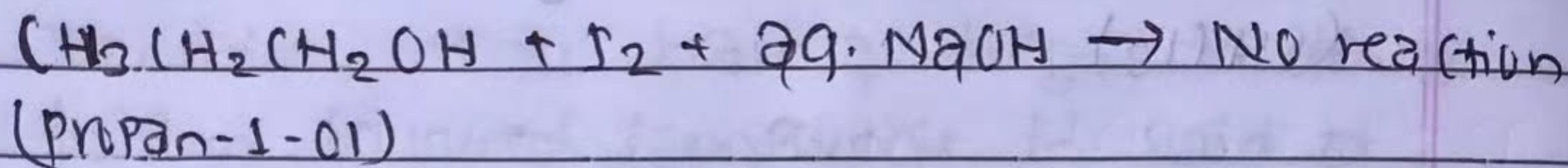


6. (b) Ans: Ethanol and Propan-1-ol, can be distinguished by iodoform test.

for Ethanol (CH_3CH_2OH) :



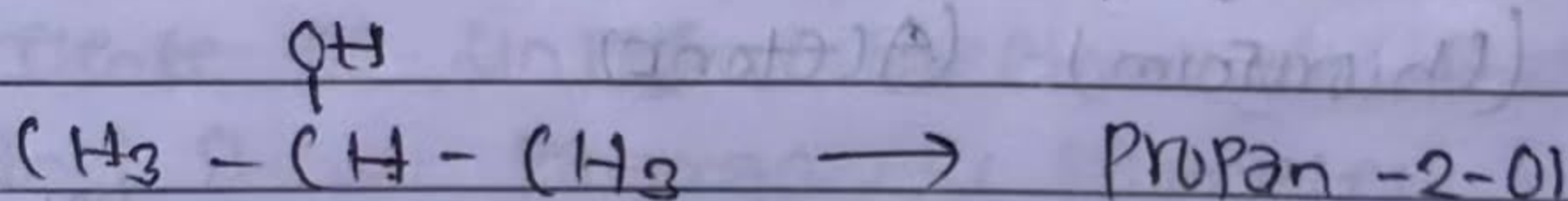
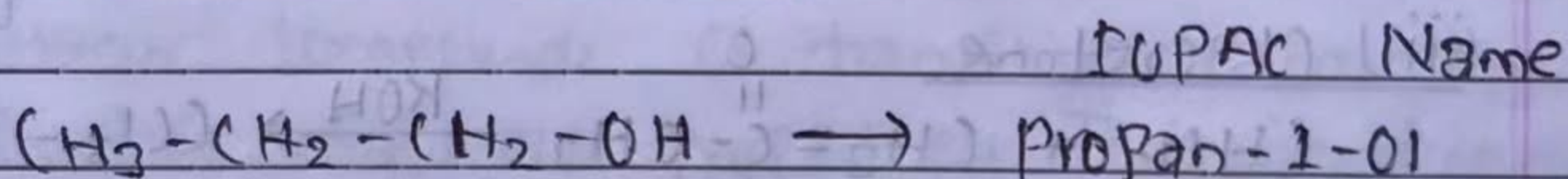
for Propan-1-ol:



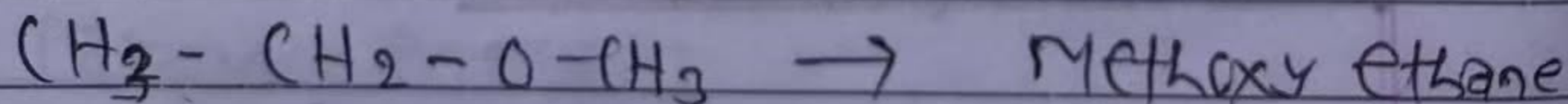
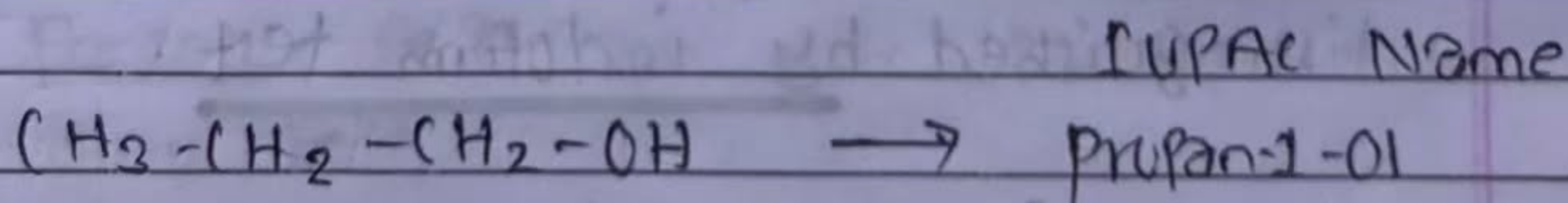
Thus, Ethanol and Propan-1-ol can be distinguished by iodoform test.

Q.N.7 (a) Ans: The formula of various positional and functional isomers of alcohols having molecular formula $\text{C}_3\text{H}_8\text{O}$ with IUPAC name are:

Positional isomers:

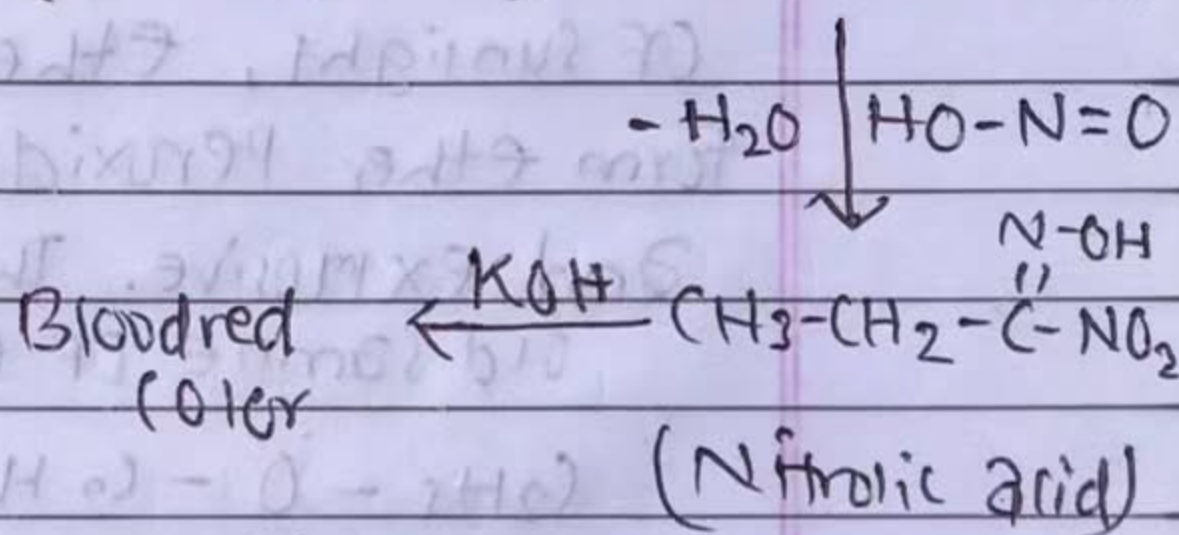
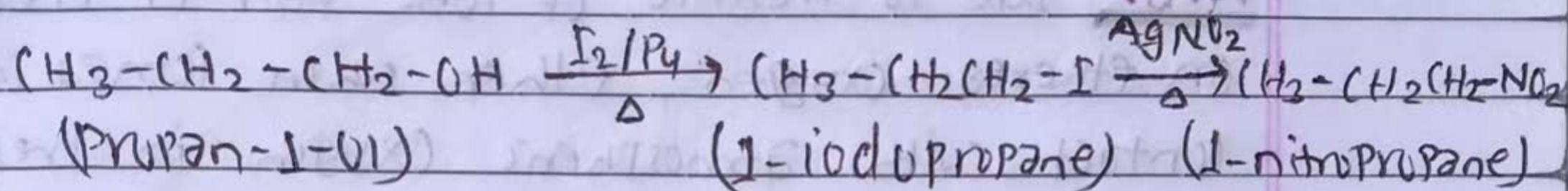


Functional isomers:

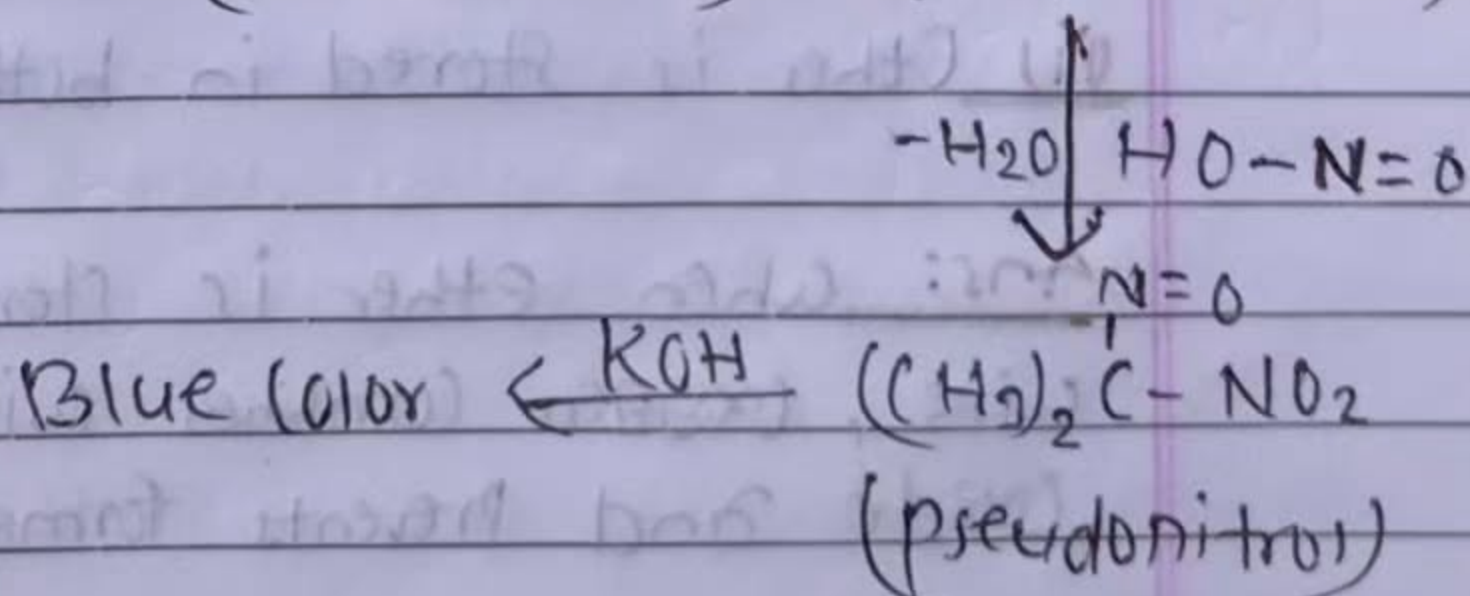
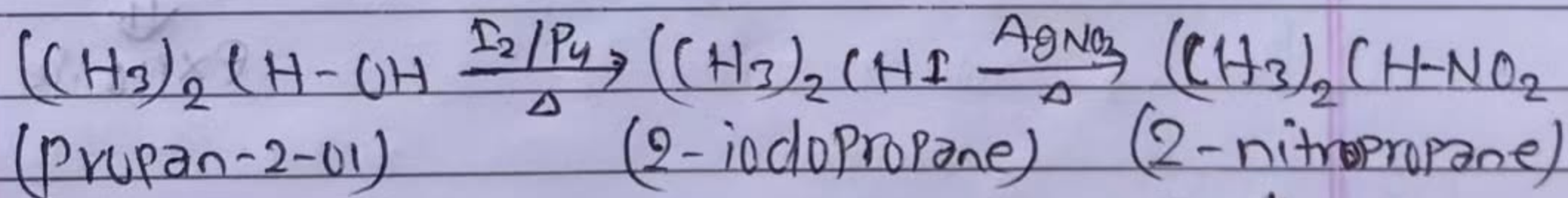


7. (b) Ans: The isomers Propan-1-ol and Propan-2-ol can be distinguished by Victor-Meyer's method.

For Propan-1-ol



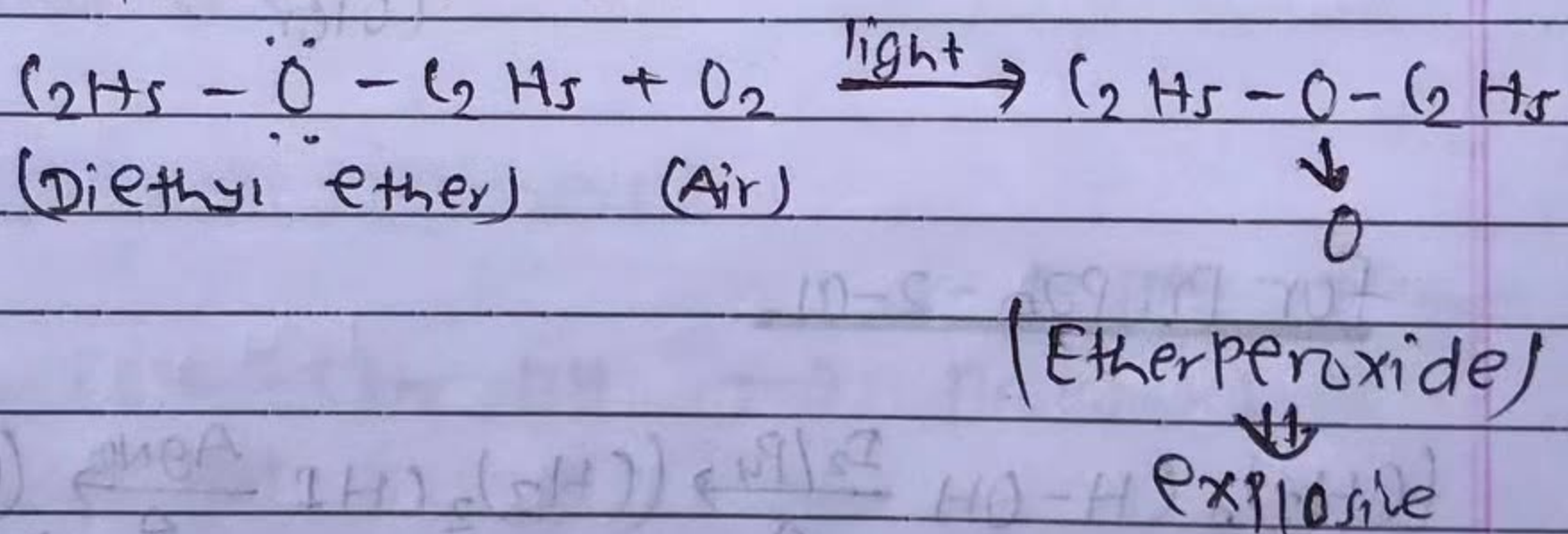
For Propan-2-ol



Q.N.8. (a) Give reason:

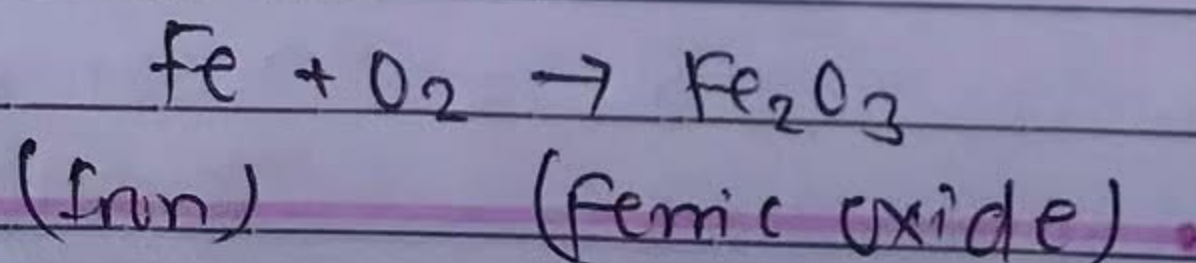
(i) It is dangerous to boil old sample of ether.

Ans: Due to presence of lone pair of electrons on etheral oxygen, ether when comes in contact with atmospheric oxygen in presence of sunlight, ether reacts with oxygen to form ether peroxide which is highly unstable and explosive. Thus, it is dangerous to boil old sample of ether.

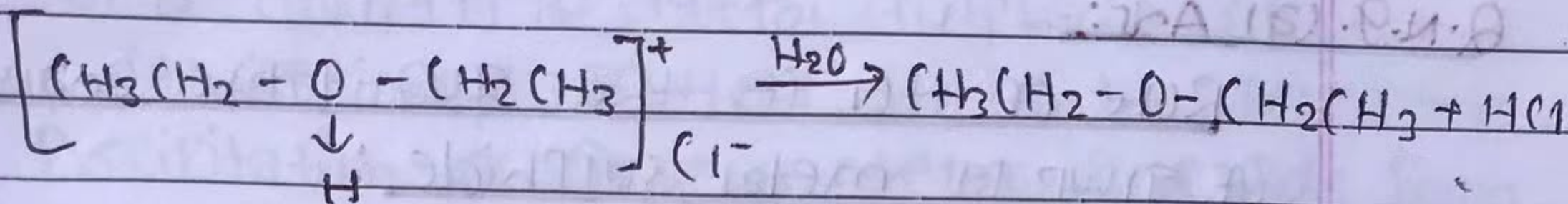


(ii) Ether is stored in bottle containing wire.

Ans: When ether is stored in bottle containing wire, oxygen combines with iron to form iron oxide and prevents formation of peroxide.



P. (b) Ans: When an Oxonium salt is dissolved in water, It decomposes to give back ether.



(Diethyl Oxonium chloride)

8. (c) Ans:

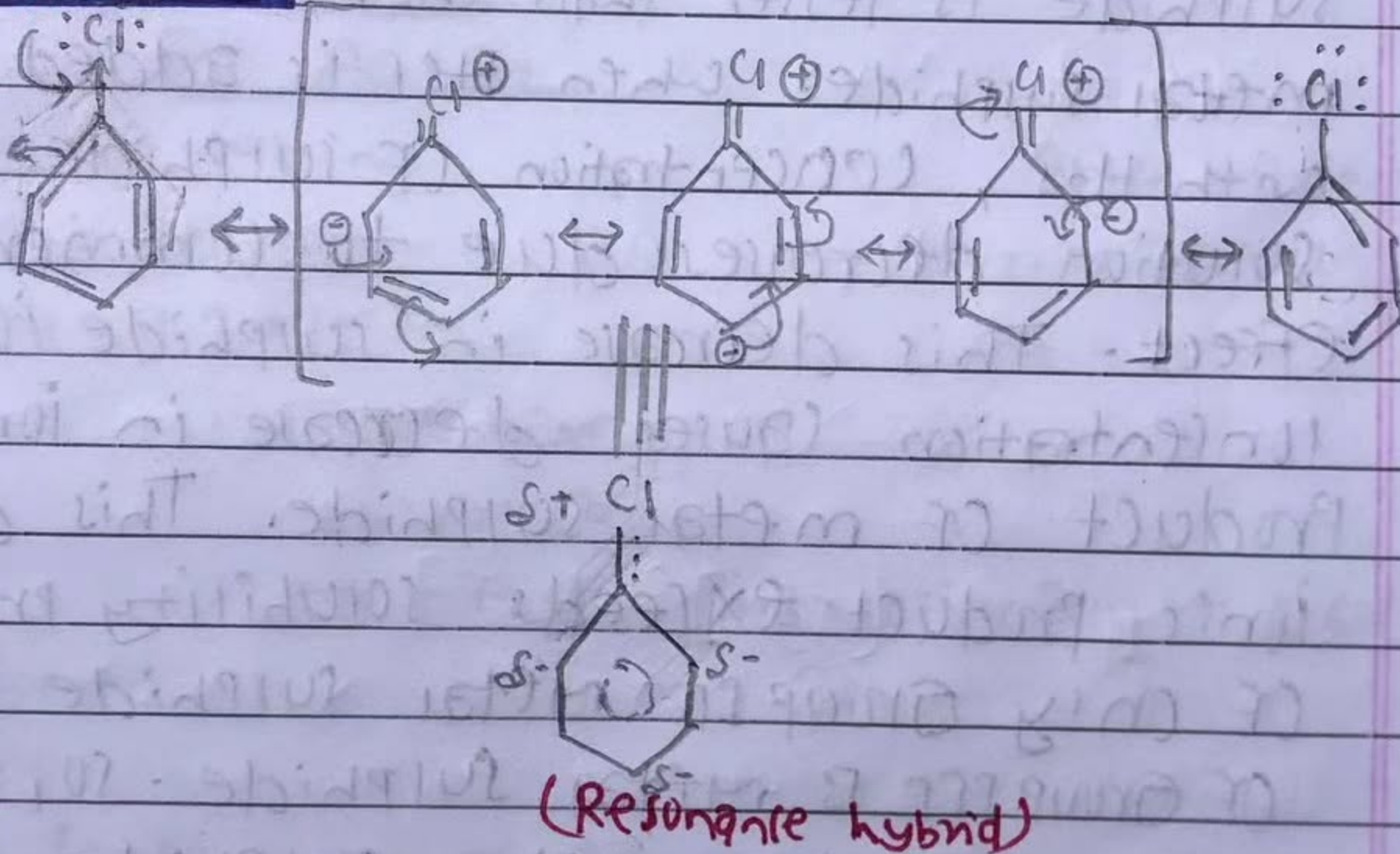


Fig: Resonating Structure of Chlorobenzene
 From above resonating structure of Chlorobenzene, It is clear that $-\text{Cl}$ group has powerful electron withdrawing ($-I$ effect) over its resonance effect. Thus, electron density decreases on benzene nucleus of Chlorobenzene as compared to benzene. Hence, Chlorobenzene is less reactive as compared to benzene towards electrophilic substitution reaction.

"Grade XII: Chemistry"

1st Term Solution

1st-Term Chemistry Paper Solution 2081

Group 'C' Long Question

Long Questions

By Pankaj Purbey

Q.N.9. (a) Ans:

Role of HCl in the qualitative analysis of Group II metal sulphide:

Both Group II & Group III B metal ions precipitate out in the form of metal sulphide. However, solubility product of Group II metal sulphide is lesser than that of Group III B metal sulphide. When HCl is added along with H_2S , concentration of sulphide ion in solution decreases due to common ion effect. This decrease in sulphide ion concentration causes decrease in ionic product of metal sulphide. This decreased ionic product exceeds solubility product of only Group II metal sulphide but not of Group III B metal sulphide. So, precipitation of only Group II metal sulphide occurs leaving Group III B metal ions in the solution.

If HCl is not added, then the concentration of sulphide ion in solution increases. This causes increase in ionic product of metal sulphide. This increased ionic product exceeds solubility product